留嘉城 Jiacheng Liu

ljc.mytcl@gmail.com (+86) 13567280711
https://tammytcl.github.io/Liu_Homepage/ Google Scholar GitHub

教育背景

上海交通大学 · 人工智能学院 2026.09 – 至今
计算机科学与技术 · 博士研究生；指导教师：张林峰教授。

山东大学 · 机电与信息工程学院 2022.09 – 2026.07
计算机科学与技术 · 本科。

研究方向

大语言模型后训练与智能体，重点关注多轮 Coding Agent 训练、长程信用分配与跨模型知识迁移；
高效多模态生成；交互式世界生成与建模。

实习经历

无尽前沿（Endless Frontier） 2026.06 – 至今
LLM 基模组 · 后训练实习生 上海

腾讯混元 · 多模态模型部 2025.07 – 2026.06
世界模型基模组 · 青云计划研究实习生 上海

大语言模型研究经历

1. Coding Agent：代码任务构建与验证

接入 SWE-Factory、SWE-rebench、FeatureBench 与 SWE-smith 等出题方法，覆盖真实 PR 修复、测试驱动的功能补全及合成缺陷修复。完成题面、仓库版本、参考补丁与测试的对齐，配套构建容器环境和评测脚本，并通过初始版本与参考解的测试对照，验证任务可解性与判分有效性。

2. 多轮 Coding Agent：高效后训练与长程监督

面向 Qwen3.5-27B→9B 的 Agent 蒸馏，实现 Rollout – Teacher – Actor 轮级异步训练，通过教师前缀复用与因果训练链合并，减少阶段等待和历史重复计算。将轻量价值头集成至冻结教师模型，利用已验证的终局结果在线学习任务成功率，并依据每轮行动及工具反馈前后的价值变化分配蒸馏权重，将任务级反馈引入过程监督。在 24 GPU 初期更新窗口中，系统有效训练吞吐较 AsyncOPD 提升 50.6%。

3. 模型知识迁移：异构蒸馏与多教师训练

参与设计与实现 ESCD，构建跨 Tokenizer 的字节级概率对齐与事件补全监督，在 Kimi 1T→Qwen 35B、Qwen 397B→30B 的异构 MoE 蒸馏中验证；其中 Kimi 配置在 PHYRD-40 物理推理基准上的平均分较最强对比基线提升 5.1 分（69.2→74.3）。开展 MOPD 多教师训练，将数学、代码与指令遵循教师的监督整合到统一学生模型，并参与面向工业代码和多模态任务的 27B 级模型合并。

以上代表工作包括 ESCD（ICLR 在投）、TRAIL、PHYRD（ICLR 在投）与 BigBang（技术报告）。

多模态研究经历

1. 高效生成模型：特征预测与自适应计算

围绕扩散模型的重复计算，研究基于历史特征演化的预测式缓存，将直接复用拓展为特征预测，并探索预测校准、阶段感知预测器、自适应计算与频域缓存，覆盖图像、视频、编辑及语音生成。代表工作包括 TaylorSeer（ICCV 2025）、SpeCa（ACM MM 2025）、LESA（CVPR 2026）与 FreqCa（NeurIPS 2026）。

2. 交互式世界生成与建模

Hunyuan-GameCraft-2：在腾讯混元参与自然语言指令与玩家动作驱动的游戏世界模型研发，基于 Unity 渲染游戏场景视频并构建训练数据；参与交互视频清洗与标注、交互记忆教师模型 SFT、Self-Forcing / DMD 少步蒸馏、强化学习优化及评测，关注动作响应与长时序生成一致性。

Deformia：参与代码驱动的交互式三维世界生成研究，基于 Unity 构建包含可形变物体的室内环境；结合扩散模型场景规划、结构感知建模与物理仿真反馈修正，通过 DOPE 评估物体放置、环境约束下的形变及动态交互响应，服务具身智能训练与评测。

以上代表工作：Hunyuan-GameCraft-2（技术报告）；Deformia（CVPR 2027 在投）。

论文与技术报告

粗体作者为本人；* 表示同等贡献，† 表示通讯作者。

[CVPR 2027 Under Review] Deformia: Agentic Simulation-Ready Indoor Worlds with Diverse Deformable Objects. Zhikai Wang, Haoran Mu, **Jiacheng Liu**, Zexuan Yan, Suning Huang, Qianzhong Chen, Jiahang Tu, Zhixuan Liang, Linfeng Zhang, Yue Ma.

[ICLR 2027 Under Review] Beyond Token Alignment: Event Completion for Cross-Tokenizer On-Policy Distillation. **Jiacheng Liu***, Song, J.*, Zhang, Q., Chen, S., Linfeng Zhang†. [论文] [代码] [报告] [SciDeriv] [PHYRD-40]
[ICLR 2027 Under Review] RolloutRemover: Scribble-Guided Object Removal via Latent Rollout Modeling. Wang, Z., Li, Y., **Jiacheng Liu**, Ma, Y., Huang, W., Liu, Q., Zhang, J.†.
[ICLR 2027 Under Review] TempoKV: Predictive KV Cache Quantization For Autoregressive Video Generation. Hu, J., Mo, W., Cai, P., **Jiacheng Liu**, Ren, J., Zhang, X., Tu, X., Linfeng Zhang†.
[ICLR 2027 Under Review] Search Low, Refine the Winner: Fast Inference through Noise-Trajectory Selection. Qin, H., Yan, Z., **Jiacheng Liu**, Peng, H., Liu, J., Tu, X., Ren, J., Zhang, X., Wang, Y., Linfeng Zhang†.
[ICLR 2027 Under Review] PHYRD: A Living Benchmark for Reproducing and Extending Theoretical Physics Research. Zhang, Q., Chen, G., Zhang, W., Wang, J., Song, J., **Jiacheng Liu**, Chen, S., Linfeng Zhang†.
[Neurocomputing Under Review] PCMamba: Parallel Convolution-Mamba Network for Medical Image Segmentation. **Jiacheng Liu**, Dong, L., Chen, B., Shi, J., Lu, W., Zhang, S., Yang, F.†, Li, D., & Zhang, L. [论文] [代码]
[TMLR 2026] A Survey of Linear Attention: Algorithm, Theory, Application, and Infrastructure. Zhang, Y., Sun, W., Sun, S., Ding, W., Xie, R., Wang, H., Chen, J., **Jiacheng Liu**, Wang, S., Zhang, Y., Huang, Y., Wang, J., Zhao, T., Han, W., Chen, Y., Zhang, K., Li, S., Xie, R., Wang, D., Chen, J., Zhang, L., Xu, C., & Wang, Y.†. [论文] [资源]
[NeurIPS 2026] Focused Forcing: Content-Aware Per-Frame KV Selection for Efficient Autoregressive Video Diffusion. Cai, P., Zhang, E., **Jiacheng Liu**, Lin, H., Zhang, R., Mo, W., Ma, Y., Zheng, S., Huang, J., Liu, D., & Zhang, L.†. [论文]
[NeurIPS 2026] Dynamic Video Generation: Shaping Video Generation Across Time and Space. Zheng, S., Huang, J., **Jiacheng Liu**, Chen, G., He, L., Lin, Y., Cai, P., & Zhang, L.†. [论文] [代码]
[NeurIPS 2026] FreqCa: Accelerating Image Generation and Editing via Frequency-Aware Caching. **Jiacheng Liu**, Cai, P., Zhou, Q., Lin, Y., Kong, D., Huang, B., Pan, Y., Xu, H., Zou, C., Tang, J., Zheng, S., & Zhang, L.†. [论文] [代码]
[NeurIPS 2026] Z-Cache: Accelerating Diffusion Transformers via Self-Reflection. Cheng, Z.*, Wang, Z.*, **Jiacheng Liu**, Tu, X., Zhou, C., Chen, J., Ma, Y., Ma, Z., Ren, J., Zou, L., & Zhang, L.†.
[ECCV 2026] LinCa: Accelerating Diffusion Models via Learnable Decomposed Feature Caching. Liu, J.*, Qin, H.*, Tu, X., **Jiacheng Liu**, Hu, J., Yan, Z., Xie, Y., Shen, K., Ren, J., Lin, Y., Zhang, X., & Zhang, L.†. [论文] [代码] [权重]
[ECCV 2026] AViTS: Adaptive Spatiotemporal Token Selection for Efficient Dynamic-Resolution Generation. Qin, H.*, Yan, Z.*, Zheng, S.*, Tu, X., **Jiacheng Liu**, Lin, Y., Zou, C., Liu, J., Cai, P., Zhang, X., Ren, J., & Zhang, L.†. [论文]
[ECCV 2026] Accelerating Diffusion Transformers with Gaussian Process Rectified Feature Cache. Shen, Z.*, Huang, R.*, Zou, C., Zheng, S., **Jiacheng Liu**, Cai, P., Shi, Z., Du, Y., Feng, L., Tu, X., Ren, J., Zhang, X., & Zhang, L.†. [论文]
[CVPR 2026 Workshop · Technical Report] Hunyuan-GameCraft-2: Instruction-following Interactive Game World Model. Junshu Tang*, **Jiacheng Liu***, Jiaqi Li*, Longhuang Wu, Haoyu Yang, Penghao Zhao, Siruis Gong, Xiang Yuan, Shuai Shao, Linfeng Zhang, Qinglin Lu†. [论文] [项目]
[Technical Report 2026] BigBang: Pursuing Open-Ended Intelligence through Self-Evolving Synthesis of Verifiable Frontier Tasks. The BigBang Team. [论文] [项目] [权重]
[进行中] TRAIL: Critic-Guided On-Policy Distillation for Multi-Turn Agents. **Jiacheng Liu***, Jingwei Song*, Siheng Chen, Linfeng Zhang†.
[BSPC 2025] FQCDM: Feature Quantization-Based Cardiac Image Diffusion Synthesis Model. Shi, J., Chen, B., **Jiacheng Liu**, Lu, W., Zhang, S., Ma, W., Yang, F.†, & Li, D. [论文]
[CVPR 2026] LESA: Learnable Stage-Aware Predictors for Diffusion Model Acceleration. Peiliang Cai*, **Jiacheng Liu***, Haowen Xu, Xinyu Wang, Chang Zou, Linfeng Zhang†. [论文] [代码]
[CVPR 2026] From Sketch to Fresco: Efficient Diffusion Transformer with Progressive Resolution. Zheng, S., Chen, G., He, L., **Jiacheng Liu**, Lin, Y., Zou, C., & Zhang, L.†. [论文]
[CVPR 2026] Beyond Fixed Formulas: Data-Driven Linear Predictor for Efficient Diffusion Models. Shen, Z.*, Huang, R.*, **Jiacheng Liu**, Zou, C., Cai, P., Zheng, S., Shi, Z., Feng, L., & Zhang, L.†. [论文] [代码]
[Survey 2026] A Survey on Cache Methods in Diffusion Models: Toward Efficient Multi-modal Generation. **Jiacheng Liu***, Wang, X.*, Lin, Y., Wang, Z., Wang, P., Cai, P., Zhou, Q., Yan, Z., Yan, Z., Shi, Z., Zou, C., Ma, Y., & Zhang, L.†. [论文] [资源]
[ICLR 2026 Oral] Let Features Decide Their Own Solvers: Hybrid Feature Caching for Diffusion Transformers. Zheng, S., Chen, G., Zhou, Q., Lin, Y., He, L., Zou, C., Cai, P., **Jiacheng Liu**, & Zhang, L.†. [论文]
[ICLR 2026] HiCache: Training-free Acceleration of Diffusion Models via Hermite Polynomial-based Feature Caching. Feng, L.*, Zheng, S.*, **Jiacheng Liu**, Wang, X., Zou, C., Ma, Y., Chen, J., Cai, P., & Zhang, L.†. [论文] [代码]
[AAAI 2026] WaveEx: Accelerating Flow Matching-based Speech Generation via Wavelet-guided Extrapolation. Liu, X.*, Gui, X.*, Ge, Z., Ge, Y., Zou, C., **Jiacheng Liu**, Niu, Z., Zheng, Q., Xu, C., Chen, X., Xiao, T., Zhu, J., & Zhang, L.†. [论文]
[AAAI 2026] Forecast then Calibrate: Feature Caching as ODE for Efficient Diffusion Transformers. Zheng, S., Feng, L., Wang, X., **Jiacheng Liu**, Zou, C., Ma, Y., Chen, J., Cai, P., & Zhang, L.†. [论文] [代码]
[ACM MM 2025] SpeCa: Accelerating Diffusion Transformers with Speculative Feature Caching. **Jiacheng Liu***, Chang Zou*, Yuanhuiyi Lyu, Fei Ren, Shaobo Wang, Kaixin Li, Linfeng Zhang†. [论文] [代码] [项目]
[BSPC 2025] PSVT: Pyramid Shifted Window based Vision Transformer for Cardiac Image Segmentation. Zhang, X., **Jiacheng Liu**, Xian, X., Chen, B., Li, D., Yang, F.†, & Zhang, L. [论文] [代码]
[ICCV 2025] From Reusing to Forecasting: Accelerating Diffusion Models with TaylorSeers. **Jiacheng Liu***, Chang Zou*, Yuanhuiyi Lyu, Junjie Chen, Linfeng Zhang†. [论文] [代码] [项目]
[Preprint 2025] Accelerate Diffusion Transformers with Feature Momentum. Fang, J., Zou, C., **Jiacheng Liu**, Lyu, Y., Hu, X., & Zhang, L.†. [论文]

奖项与荣誉

创业实践类与研究创新类特长奖一等奖、优秀学生一等奖、校三好学生。